

Day 2 – Dataset Analysis & Classification

Personality Dataset &
Loan Dataset



Personality Dataset – Phase 1 & 2

- Data loaded from `personality_dataset.csv`
- Checked shape, columns, missing values
- Dropped rows with missing values
- Features: mix of categorical & numerical
- Preprocessing:
 - - One-Hot Encoding for categorical features
 - - Label Encoding for target (Personality)



Personality Dataset – Phase 3 & 4

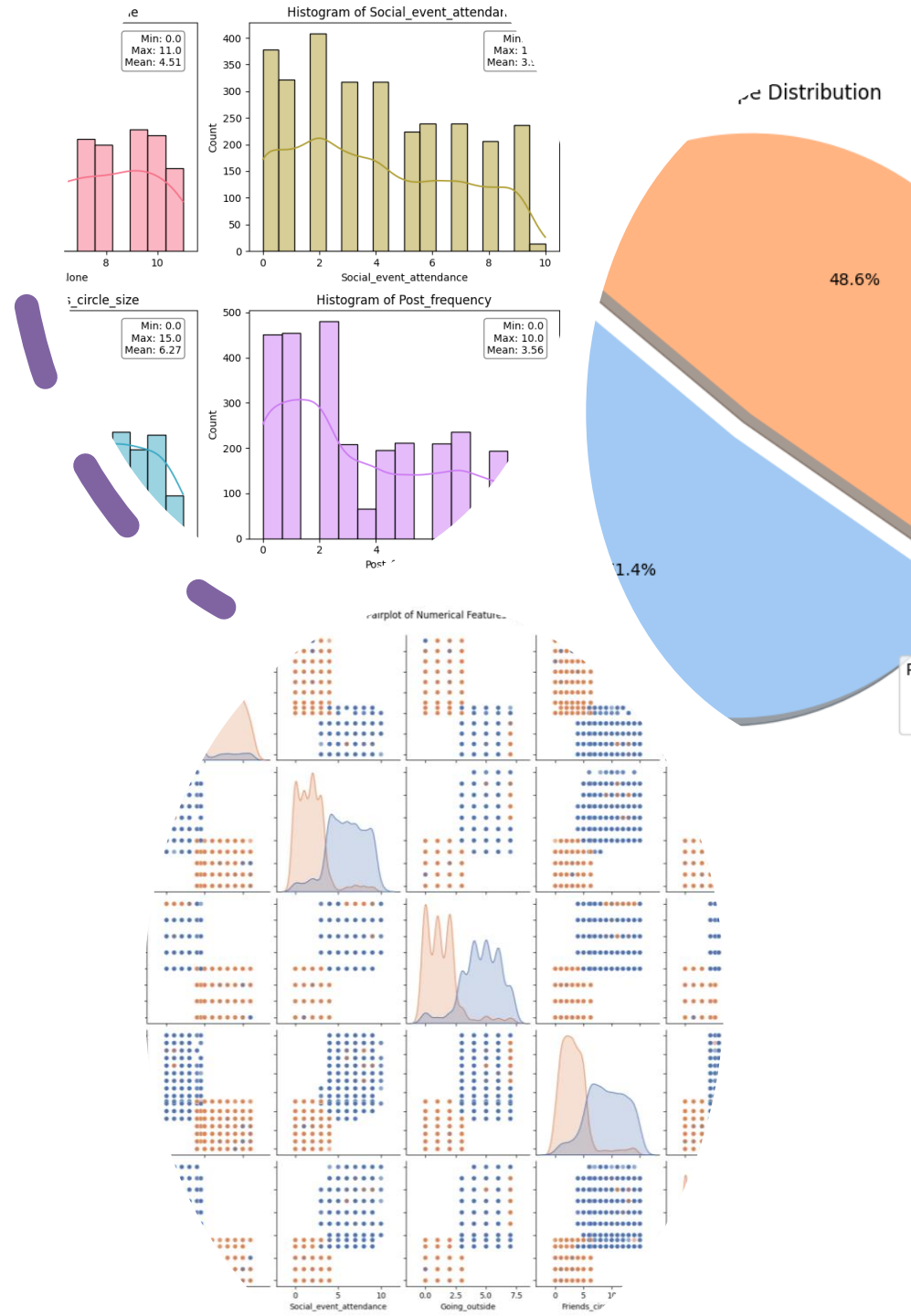
Visualizations:

- Bar plots (categorical features)
- Box plots (numerical features)
- - Distribution plots of traits
- Model Preparation:
- X = Features, y = Personality

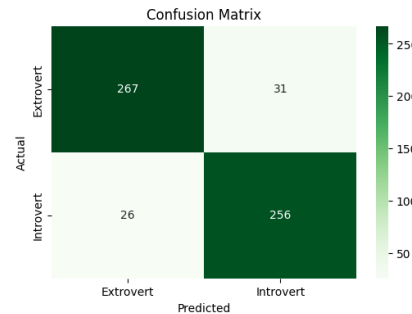
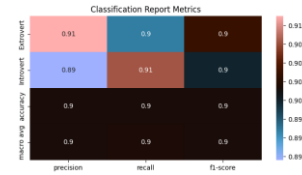
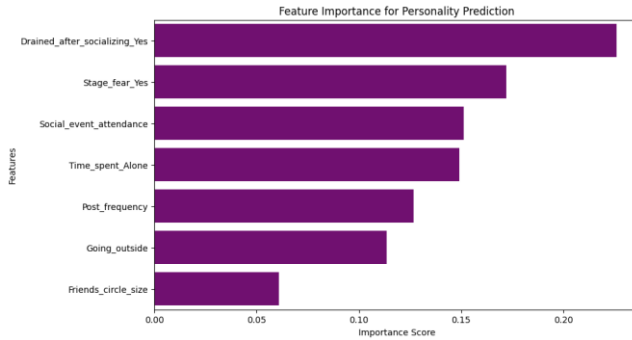
Train-Test Split

Model:

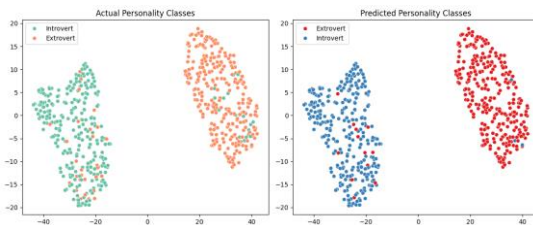
- Random Forest Classifier (100 trees)



Personality Dataset – Phase 5 & 6



- Evaluation:
- - Confusion Matrix
- - Classification Report (precision, recall, F1)
- Feature Importance → key predictive variables
- t-SNE:
- - Reduced high-dimensional data into 2D
- - Clusters show separability, overlap shows complexity



Loan Dataset – Phase 1 & 2

- Dataset: 598 records, 13 features
- 7 categorical + 6 numerical

Key findings: majority male, married, graduate →
most loans approved

Preprocessing:

Label Encoding

- - Missing values handled with median imputation

Loan Dataset – Phase 3 & 4

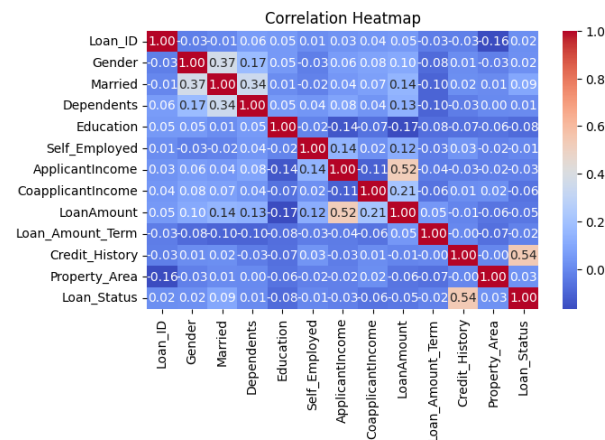
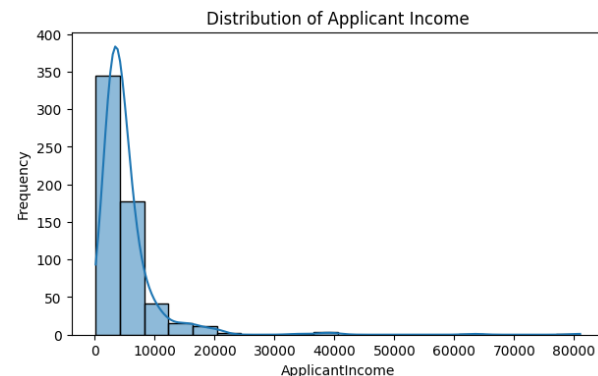
Visualization:

Correlation heatmap

- Credit History → strongest link with Loan Status
- LoanAmount moderately correlated with incomes
- Dependents & Loan Term → weak predictors

Data Split:

- 60% training, 40% testing
- -Standardized features



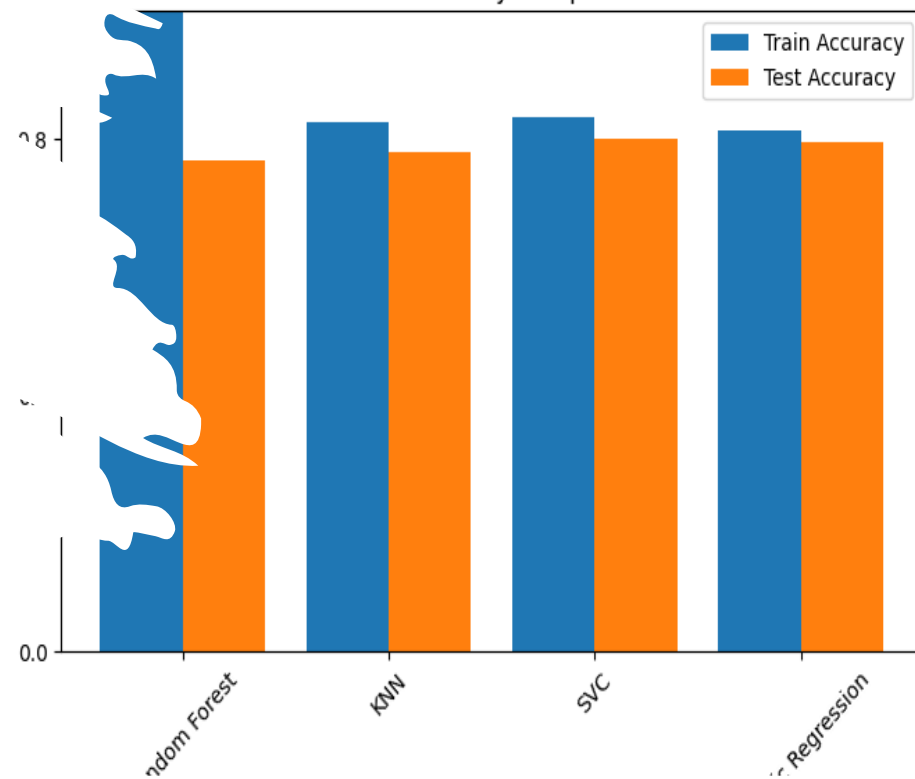
Loan Dataset – Phase 5 & 6

Models Trained:

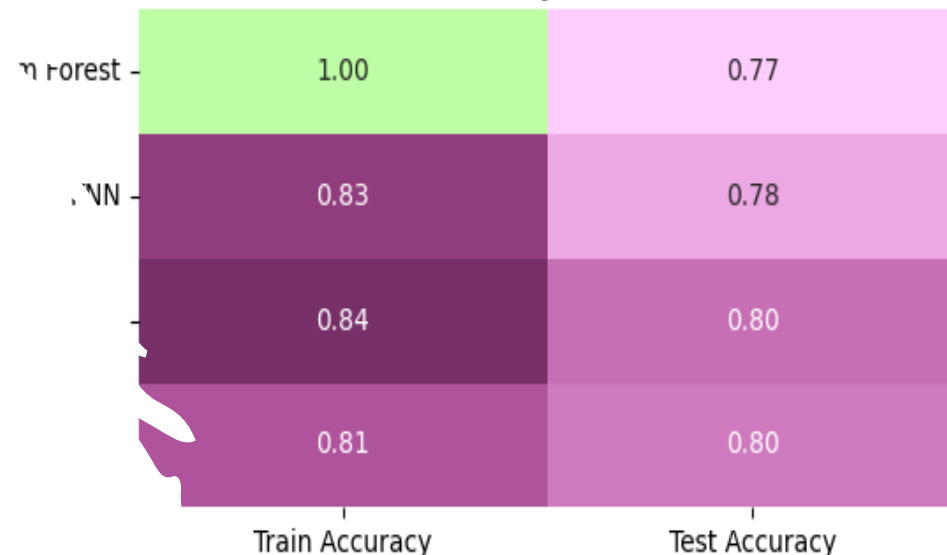
- - Random Forest → 100% train acc (overfit)
- - KNN → 82.7% train acc
- - SVC → 83.5% train acc
- - Logistic Regression → 81.3% train acc

Model Evaluation:

- - Best test accuracy: SVC = 80%
- - Logistic Regression = 79.6%
- - KNN = 74.6%
- - Random Forest = overfit



Model Accuracy (Train vs Test)





Final Conclusions

Personality Dataset:

- - Random Forest effective, with t-SNE visualization

Loan Dataset:

- - Best model: Support Vector Classifier (80%)

Both analyses:

- - Importance of preprocessing, encoding, visualization
- - Model comparison & evaluation is key